



SREEPARVATHY

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PROFILE SUMMARY

Bioinformatics researcher passionate about emerging technologies at the intersection of computational biology and medicine, including multi-omics, spatial transcriptomics, and LLMs. Driven by an endless appetite for daily learning and problem-solving, I am seeking a PhD opportunity where I can apply novel computational methods across any specialized biological niche to directly improve human or animal lives.

EDUCATION

- **Master of Science (M.Sc.) – Bioinformatics (2023 - 2025)** **CGPA: 9.53/10**
Garden City University, Bengaluru, Karnataka
- **Bachelor of Science (B.Sc.) – Microbiology (2019 - 2022)** **CCPA: 7.53/10**
Indira Gandhi College of Arts and Science, Ernakulam, Kerala

EXPERIENCE

- **Independent Bioinformatics & Computational Biology Collaborator**
Bengaluru | Sep 2026 – present
 - Co-developing and reviewing hands-on, containerized pipelines in Python, Nextflow, and Conda for targeted genomic analysis, encompassing short variant calling, CNV segmentation, gene fusion detection, and clinical LLM workflows.
- **Bioinformatics Intern**
Sageloom Analytics, Bengaluru | Apr 2026 – Jul 2026
 - Performed exploratory data analysis (EDA) and systematic literature mining on Indian Rare Genetic Disease (RGD) cohorts, specializing in neuromuscular disorders.
 - Built a high-throughput, parallelized Python/Linux tool using $O(\log n)$ bisect insertion and RAM disk streaming to generate custom spiked-in VCFs for workflow validation.
 - Assisted in the optimization, documentation, and production management of clinical NGS pipelines within a Python-based Linux environment.
- **Research Associate Intern / Research Associate**
Sangene Biotech, Bengaluru | Jan 2025 – Sept 2025
 - Started as an intern, transitioning to Research Associate in Bioinformatics Quality Control from July 2025 to September 2025.
 - Mentored students in bioinformatics, covering NGS data analysis, RNA-seq interpretation, and project development, enhancing their technical skills.
 - Collaborated with microbiology and molecular biology teams to design experiments.
 - Created standardized workflows and educational materials for omics data analysis, boosting team efficiency.

- **Research Intern**

Garden City University, Bengaluru | Oct 2023 – 2026

- **Antimicrobial Drug Design for *Acinetobacter baumannii***

Collaboration with Dr. Bruno Iopes, Teesside University, UK.

- Annotated the complete genome using Prokka to identify potential drug targets.
- Performed structural analysis and virtual screening for novel antibiotics.

- **Neural Informatics-Based Antidepressant Development**

Collaboration with Dr. Rammohan Shukla, University of Wyoming/University of Toledo, USA.

- Investigated the 80s ribosomal structure as a target for depression therapeutics.
- Applied molecular docking and dynamics simulations for lead optimization.

- **Research Intern**

Center for Research on Molecular and Applied Science (CRMAS), Thiruvananthapuram |

Dec 2021 - Jan 2022

- **Comparative Analysis of Fluorochromes in DNA Damage Assessment**

Supervised by Dr. Rajesh Ramachandran

- Evaluated efficiency of ethidium bromide, acridine orange and fluorescein diacetate in comet assays
- Quantified oxidative stress-induced DNA damage in L929 cell lines using fluorescence microscopy
- Results contributed to BSc Microbiology dissertation at Mahatma Gandhi University

SKILLS

Technical Skills:

- **Languages:** Python (NumPy, Pandas, Biopython), R (tidyverse, Bioconductor), Bash
- **Data Science:** Scikit-learn, TensorFlow, Jupyter Notebooks
- **Workflow Tools:** Snakemake, Nextflow
- **Databases:** SQL, UCSC Genome Browser, ENSEMBL

Bioinformatics Tools:

- **Genomics:** BWA, Bowtie2, GATK, SAMtools, DESeq2
- **Structural Biology:** PyMOL, ChimeraX, AutoDock, GROMACS
- **Machine Learning:** Logistic Regression, KNN, Random Forest, XGBoost (applied to genomics)
- **Visualization:** ggplot2, Matplotlib, Seaborn, Cytoscape

Wet Lab & Analysis Techniques:

- **Microbiology:** Staining Techniques, Biochemical Testing, Cell Culture, Water Testing
- **DNA Analysis:** Comet Assay, Gel Electrophoresis Documentation and Analysis

Soft Skills:

- **Research & Analysis:** Critical Thinking, Data Interpretation, Experimental Design
- **Collaboration:** Teamwork, Interdisciplinary Communication
- **Project Management:** Time Management, Adaptability, Problem-Solving

PROJECTS

- **Whole Exome Variant Discovery** and Interpretation in a Human Cystic Fibrosis Patient Sample (2026).
- **Bulk RNA-Seq Differential Expression Analysis** of GPR34 Knockout in Human Microglia-Like Cells Using DESeq2 (2026).
- **Single-Cell and Single-Nucleus RNA-seq Analysis** of Public Octopus Brain Data (2026).
- **Variant Calling and Functional Annotation** of Erythromycin Resistance Mutations in *Deinococcus radiodurans* (2026).
- **SPIDEE: A Hybrid Bioinformatics and Machine Learning Tool** for Rapid Antimicrobial Resistance Prediction from Genetic Sequences (2025).
- **Genome-Guided Antimicrobial Drug Discovery** Against *Acinetobacter baumannii* (2023- 2025)
- Computational Design of Ribosomal-Targeted Antidepressants (2024-2026)
- Immunoinformatics-driven design and construction of a **multi-epitope peptide vaccine** candidate against rabies virus (2024).
- **Comparative Analysis of Machine Learning Models:** Logistic Regression, KNN, Random Forest, and XGBoost for Lung Cancer Classification (2024).
- **Comparative analysis** on the effect of different Fluorochromes on analysis of DNA damages by comet assay (2022)

PUBLICATIONS

- Tirumalasetty, D., Seerla, V., Akshay, P. P., Dsouza, P. M., Nandish, V., **Sreeparvathy**, Anieth Noel, A., Subbaiah, V. K., Prasad, L. A. R. C., & Lopes, B. S. (2026). **Network analysis of polA in Acinetobacter baumannii International Clone 2 and computational evaluation of nodulisporone and nodulisporol as potential inhibitors.** *APMIS*, 134(1), Article e70149. <https://doi.org/10.1111/apm.70149>

AWARDS AND HONORS

- **Best in Academics** - Garden City University Cream Awards 2024
- **Participation for Poster Presentation** - RAMIAH University trends in multidisciplinary research (Department of Biotechnology of India)

CERTIFICATIONS

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| • Genomic Data Science (specialization)
<i>Coursera</i> | 2023 |
| • Epigenetic Control of Gene Expression
<i>Coursera</i> | 2024 |
| • Finding Hidden Messages in DNA (Bioinformatics I)
<i>Coursera</i> | 2023 |
| • Big Data, Genes, and Medicine
<i>Coursera</i> | 2024 |
| • Drug Development
<i>Coursera</i> | 2024 |
| • Hands-on Introduction to Linux Commands and Shell Scripting
<i>Coursera</i> | 2024 |
| • Introduction to R Programming for Data Science
<i>Coursera</i> | 2024 |
| • 2-days Hands-on Workshop on “Bioinformatics Data Analysis”
<i>Institute of Bioinformatics and Applied Biotechnology</i> | 2024 |

REFERENCES

- Dr. L. A. Rama Chandra Prasad,,
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